A Physics-Inspired and Data-Driven Nonlinear Hyperspectral Unmixing Model via a Recursive Interaction Structure

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***Abstract*—Hyperspectral unmixing is a fundamental task for fine-grained material characterization, aiming to decompose mixed pixels into pure endmember spectra and their corresponding abundances. However, traditional linear mixing models (LMMs) are inherently limited by neglecting complex nonlinear scattering effects. To address this, existing multilinear models typically rely on physics-driven formulations based on prior knowledge, such as scattering probabilities. In this work, we propose a novel data-driven modeling paradigm with a physics-inspired recursive interaction structure. By explicitly constructing high-order interactions between linear components and latent features through learnable weights at each layer, the model systematically captures composite nonlinearities. Furthermore, a learnable scalar parameter is introduced to dynamically gate the contribution of nonlinear components, significantly enhancing the model's adaptive representation capability. To implement the proposed model, a concise and efficient dual-stream network architecture is developed. Specifically, the abundance estimation stream integrates multilayer perceptrons (MLPs), convolutional neural networks (CNNs), and attention mechanisms to effectively extract joint spatial–spectral features. Meanwhile, an independent endmember stream is designed to distill pure spectral signatures from a global perspective. Combined with a pretraining strategy, the proposed framework significantly reduces computational complexity while achieving competitive unmixing accuracy, thereby striking a superior balance between performance and efficiency.**

***Index Terms*—Hyperspectral unmixing, nonlinear mixture model, physics-inspired neural network, recursive interaction structure.**

I. Introduction

HYPERSPECTRAL imaging acquires hundreds of contiguous narrow spectral bands over the same scene, forming a 3-D data cube. Its high spectral resolution allows distinguishing differences in material characteristics, facilitating accurate identification and quantitative analysis across fields including agricultural monitoring [1], [2], [3], environmental surveillance [4], medical diagnostics [5], and land-use classification [6]. However, this spectral advantage is typically achieved at the expense of spatial resolution, causing a single pixel to cover multiple materials, such as soil, vegetation, or buildings, resulting in mixed pixels. The observed spectra of such pixels are mixtures of different material signatures, limiting their use in classification or inversion. Hence, spectral unmixing is required to reveal the underlying material composition and distribution within each pixel.

Hyperspectral unmixing decomposes mixed pixels in hyperspectral images into pure material spectral signatures, known as endmembers, and their corresponding proportions, termed abundances [7], [8]. Unmixing methods are generally classified into linear and nonlinear mixing models (NLMMs) based on the complexity of light–matter interactions within the scene. Linear mixing models (LMMs) assume that observed spectra are linear combinations of endmembers weighted by their abundances [9]. Endmembers and abundances are typically estimated by minimizing the difference between the reconstructed and observed spectra [10], [11]. In contrast, NLMMs assume a nonlinear relationship between endmembers and abundances, accounting for multiple light–material interactions. These include bilinear models [12], [13], [14], multilinear models [15], [16], and neural network-based approaches designed to capture complex nonlinear mixing effects [17], [18].

A. Classic Unmixing

Traditional linear unmixing approaches encompass a wide range of methods, among which geometric strategies are widely studied, including endmember extraction using vertex component analysis (VCA) [19] followed by abundance estimation via fully constrained least squares (FCLSs) [10], as well as mix source separation techniques based on nonnegative matrix factorization [11], [20], [21]. These methods perform well in many scenarios and have laid the foundation for subsequent research.

In real-world conditions, the interaction between light and materials is often more complex than the linear assumption suggests, and nonlinear effects mainly arise from two typical situations. The first is intimate mixing at the microscopic scale, commonly observed in granular media such as sand, mineral mixtures, or planetary surfaces, where photons undergo multiple scattering. A representative model for this case is the Hapke model derived from radiative transfer theory [22]. The second is scene mixing at the macroscopic scale, which results from hierarchical structures among different materials, becoming particularly significant in vegetation canopies or soil–vegetation transition zones. To address these complexities, a variety of data-driven nonlinear models have been proposed. A representative class is bilinear models, which

Received 11 January 2026; revised 21 April 2026 and 18 August 2026; accepted 7 September 2026. Date of publication 10 September 2026; date of current version 18 September 2026. *(Corresponding author: Hongjuan Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3732509

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extend the linear framework by modeling second-order photon interactions between materials. Typical examples include the Fan model [12] and the generalized bilinear model (GBM) [13], the latter allowing flexible adjustment of nonlinearity through parameterization. Another approach is the polynomial postnonlinear model (PPNM) [14], which represents spectral distortions using polynomial functions. Although these methods enhance the characterization of nonlinear effects, they primarily consider second-order interactions and thus fail to capture higher-order scattering. To overcome this limitation, the multilinear mixing model (MLM) [15] extends PPNM by incorporating the probability of high-order photon interactions, thereby accounting for scattering of theoretically infinite order. Nevertheless, MLM assumes identical scattering strength across all spectral bands, which contradicts the wavelength-dependent nature of spectral mixing. The extended multilinear mixing model (EMLM) [16] vectorizes the interaction parameters, enabling band-specific nonlinear intensities and achieving a closer alignment with physical reality. Building on this framework, the coarse-to-fine unmixing model [23] further enhances performance by exploiting spatial information: coarse unmixing is first performed via superpixel segmentation, and the results are subsequently refined under the guidance of EMLM.

B. Deep Learning-Based Unmixing

Autoencoder-based frameworks are among the most widely used deep unmixing methods [24], [25], [26], [27]. Notably, the DAEN [28] was among the pioneering works to provide a mathematical explanation for utilizing deep autoencoders in unmixing, effectively interpreting the encoder as a feature extractor for abundances and the decoder as a physical mixing model. Typically, the encoder estimates abundances, while the decoder performs reconstruction. When the decoder is implemented as a single fully connected layer, its weights can be interpreted as endmembers, which corresponds to the LMMs [24], [25]. To enhance nonlinear modeling capability, some studies embed classical NLMMs, such as PPNM, into autoencoders by adding extra layers in the decoder to characterize second-order scattering effects [26], [27], [29], [30]. Further research employs multilayer networks to approximate complex nonlinear processes [17], [31]. Other studies integrate MLM expressions with neural networks, combining model priors with deep learning to better represent nonlinear mixing scenarios [18], [32], [33]. Another line of work, distinct from neural-network-based approaches, leverages kernel mapping to achieve nonlinear unmixing in high-dimensional spaces, such as the ACK method proposed by Liu et al. [34], which incorporates ADMM-based optimization for improved physical interpretability. Overall, these methods have improved the characterization of nonlinear effects to some extent, but there remains room for further development in their model design. In fact, recent innovations in network architectures for linear unmixing, such as refined spatial feature modeling and sophisticated structural designs, offer valuable insights and guidance for the development of nonlinear models.

In terms of spatial modeling, early approaches employed convolutional neural networks (CNNs) [17] or long short-term memory (LSTM) networks [35] to jointly capture spectral–spatial features. While these methods can exploit local neighborhood information, they remain limited in modeling long-range dependencies across pixels. To address this limitation, attention-based networks have been increasingly introduced. For example, UST-Net [36] applies self-attention to entire images to alleviate patch inconsistency, while MSGA-CD [37] and MAT-Net [38] further enhance feature fusion through multiscale and attention mechanisms. More recently, SRViT [39] employs a self-supervised relation-aware vision transformer to improve the modeling of complex spatial relationships. From the perspective of model architecture, conventional autoencoder-based methods typically rely on decoder weights to jointly represent endmembers and abundances, which may lead to training instability and limited generalization. To mitigate this issue, dual-stream architectures [40], [41] have been proposed to model endmembers and abundances in separate branches and jointly optimize them during reconstruction. Building on this idea, EOT-Net [42] integrates a transformer framework with endmember-guided attention, further improving estimation robustness. Despite these advances in spatial feature modeling and architectural design, effectively leveraging them for nonlinear spectral mixing remains challenging. Some studies incorporate GBMs [43], but their flexibility and applicability are still limited. Therefore, developing more flexible and efficient nonlinear modeling approaches that fully exploit spatial–spectral representations remains an open problem.

C. Motivations and Contributions

In summary, existing studies have achieved significant progress in nonlinear extension, spatial feature modeling, and network structure decoupling. However, these efforts are often focused on specific aspects, lacking a unified framework that systematically integrates these advantages. To address this, the proposed work adopts a more holistic design, which not only combines the architectural benefits of dual-stream decoupling and spectral–spatial feature modeling but also introduces a novel nonlinear paradigm: a recursive interaction structure. This structure progressively generates higher-order interactive features in a data-driven manner, simultaneously addressing feature decoupling and complex nonlinear modeling within a unified framework, thereby establishing an unmixing approach distinct from existing methods. Building upon these motivations and to address the aforementioned challenges, the main contributions of this work can be summarized as follows.

1) In the abundance estimation module, we develop an encoder network integrated with a convolutional block attention module (CBAM). The design enables the adaptive enhancement and deep fusion of spatial–spectral features, significantly improving the precision of material distribution characterization in complex scenes.
2) In the endmember estimation module, we introduce an independent network stream that is decoupled from the abundance estimation process. By isolating endmember learning, the model mitigates the error propagation typically found in joint optimization frameworks, allowing for the extraction of more pure spectral signatures.

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3) In the spectral reconstruction stage, we design a physics-inspired recursive interaction structure, whose parameters are learned in a data-driven manner. By iteratively performing high-order coupling between the linear base and progressively refined feature representations, the model approximates multiorder scattering effects without requiring explicit physical parameters. This recursive design ensures that the nonlinear components are both mathematically grounded and highly adaptive to diverse mixing scenarios.

The remainder of this work is organized as follows. Section II provides a comprehensive review of classic unmixing models, including linear, nonlinear, and multilinear approaches. The detailed network architecture of the proposed method is shown in Section III. Section IV demonstrates the experimental results and analysis of the proposed method on both synthetic and real datasets. Finally, Section V summarizes the work.

II. Related Works

A. Linear Mixture Models

The LMM serves as the foundation of hyperspectral unmixing, which assumes that each pixel can be represented as a linear combination of endmembers weighted by their abundances. For a hyperspectral image with L spectral bands and R endmembers, the spectral observation of the ith pixel $\boldsymbol{y}_i \in \mathbb{R}^{L\times 1}$ is formulated as

$$\boldsymbol{y}_i = \boldsymbol{M}\boldsymbol{a}_i + \boldsymbol{n}_i \tag{1}$$

where $\boldsymbol{M} \in \mathbb{R}^{L\times R}$ denotes the endmember matrix, $\boldsymbol{a}_i \in \mathbb{R}^{R\times 1}$ is the abundance vector, and $\boldsymbol{n}_i$ represents noise or modeling error. The abundance vector $\boldsymbol{a}_i$ is subject to the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC), i.e., $\boldsymbol{a}_i \succeq 0$ and $\mathbf{1}^\top \boldsymbol{a}_i = 1$.

B. Nonlinear Mixing Model

1) Multilinear Mixing Model: The MLM [15] extends the traditional linear model by incorporating a multiple-scattering mechanism to characterize light interactions among different materials. The model is based on the assumption that, after a photon initially interacts with a material, it may be scattered to other materials with a certain probability p_i, where the probability of subsequent interaction with a given material is proportional to its abundance. Based on these assumptions, the observed spectrum of the ith pixel can be expressed as

$$\begin{aligned}\boldsymbol{y}_i &= (1-p_i)\sum_{j=1}^{R}\boldsymbol{m}_j a_{ij} + (1-p_i)\,p_i\sum_{j,k=1}^{R}\left(\boldsymbol{m}_j \odot \boldsymbol{m}_k\right) a_{ji}a_{ki} \\ &\quad + (1-p_i)\,p_i^2\sum_{j,k,l=1}^{R}\left(\boldsymbol{m}_j \odot \boldsymbol{m}_k \odot \boldsymbol{m}_l\right) a_{ji}a_{ki}a_{li} + \ldots, \\ &= (1-p_i)\,\boldsymbol{z}_i + p_i\boldsymbol{z}_i \odot \boldsymbol{y}_i \end{aligned} \tag{2}$$

where $\boldsymbol{z}_i = \sum_{j=1}^{R}\boldsymbol{m}_j a_{ij}$ corresponds to the linear mixing component, $\odot$ denotes the Hadamard product, and p_i is a scalar parameter that quantifies the probability of photon scattering.

2) Deep Unmixing Models: Recent studies have explored deep learning-based unmixing frameworks built upon traditional physics-based NLMMs, where physical mixing mechanisms are integrated with neural network architectures to enable parameterized modeling of nonlinear mixing processes. A typical strategy is to embed nonlinear interaction terms into a learnable modeling structure. Some approaches extend classical nonlinear models, such as bilinear or polynomial formulations, into deep architectures, which can be generally expressed as

$$\boldsymbol{y}_i = \boldsymbol{M}\boldsymbol{a}_i + \mathcal{F}_\theta(\boldsymbol{M}, \boldsymbol{a}_i) \tag{3}$$

where $\mathcal{F}_\theta(\cdot)$ denotes a learnable nonlinear mapping parameterized by a neural network. In many cases, $\mathcal{F}_\theta$ is constructed based on predefined interaction forms, such as bilinear or low-order polynomial combinations.

In addition to low-order interaction modeling, some works incorporate multiple-scattering mechanisms into deep architectures by drawing inspiration from recursive formulations such as the MLM in (2). In these approaches, the reconstruction process is typically formulated based on the MLM expression, with associated parameters learned via neural networks [18], [32], [33], or the recursive interaction process is approximated by specifically designed network structures [44], [45].

III. Proposed Method

A. Overview

In this section, we propose a nonlinear hyperspectral unmixing method built upon a novel formulation for modeling nonlinear interactions. The framework consists of three key modules: abundance estimation, endmember estimation, and spectral reconstruction, as illustrated in Fig. 1. The abundance estimation module extracts spectral and spatial features through two separate branches. The resulting features are concatenated and refined by a CBAM, which sequentially applies channel-wise and spatial attention to focus on the most informative spectral bands and spatial regions, thereby improving abundance estimation accuracy. The endmember estimation module first encodes pixel-level features using a series of 1×1 convolutional layers, then aggregates global information via adaptive average pooling. The pooled features are mapped to the endmember matrix through a multilayer perceptron (MLP) and normalized to produce the final endmembers. The spectral reconstruction module comprises both linear and nonlinear components, with the nonlinear component leveraging a cross-network design to recursively model high-order interactions between the linear reconstruction and evolving feature representations, capturing complex nonlinear mixing effects. Finally, a two-stage training strategy is employed to stabilize the optimization process and progressively enhance overall network performance.

B. Abundance Estimation Module

The abundance estimation module is designed to generate accurate abundance maps by jointly modeling spectral and spatial dependencies. Given a hyperspectral image $\boldsymbol{X} \in \mathbb{R}^{L\times H\times W}$, where L denotes the number of spectral bands and H, W

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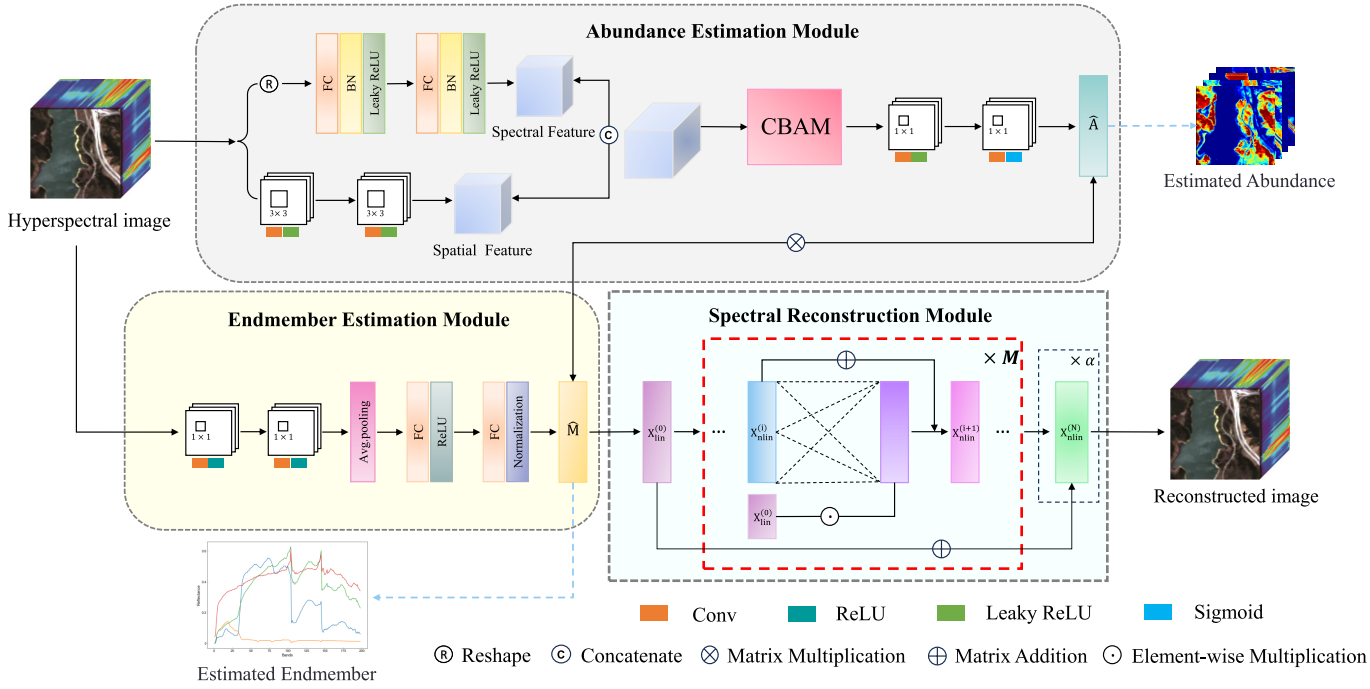


Fig. 1. Overall architecture of the proposed nonlinear hyperspectral unmixing model. The framework consists of three modules: abundance estimation, endmember estimation, and spectral reconstruction. BN denotes batch normalization.

represent the spatial dimensions, the module comprises three main stages: dual-branch feature extraction, attention-based optimization, and abundance generation.

1) Dual-Branch Feature Extraction: To effectively capture complementary spectral and spatial information, the input feature is processed through two parallel pathways. The spectral branch, denoted as $g_{\text{mlp}}(\cdot)$, first flattens the spatial dimensions to form a matrix of size $L \times N$, where $N = H \times W$ denotes the total number of pixels and each row corresponds to a pixel's spectral vector. This matrix is then passed through two consecutive fully connected layers, projecting each spectral vector into a d-dimensional feature space. Finally, the resulting feature matrix is reshaped to restore the original spatial grid, producing the spectral feature map. The overall operation can be concisely expressed as

$$\boldsymbol{F}_{\text{spec}} = g_{\text{mlp}}(\boldsymbol{X}) \in \mathbb{R}^{d\times H\times W}. \tag{4}$$

In parallel, the spatial branch extracts spatial features by applying two successive 3×3 convolutional layers to the input, capturing local spatial correlations among neighboring pixels. The operation of the spatial branch can be formulated as

$$\boldsymbol{F}_{\text{spat}} = g_{\text{cnn}}(\boldsymbol{X}) \in \mathbb{R}^{d\times H\times W}. \tag{5}$$

2) Attention-Based Optimization: The extracted spectral and spatial features are concatenated along the channel dimension and fed into a CBAM [46] to adaptively refine feature representations

$$\boldsymbol{F}_{\text{cbam}} = \text{CBAM}\left(\text{Concat}\left[\boldsymbol{F}_{\text{spec}}, \boldsymbol{F}_{\text{spat}}\right]\right) \in \mathbb{R}^{2d\times H\times W}. \tag{6}$$

The detailed structure of CBAM is illustrated in Fig. 2. It sequentially applies two complementary attention mechanisms. The channel attention submodule adaptively reweights

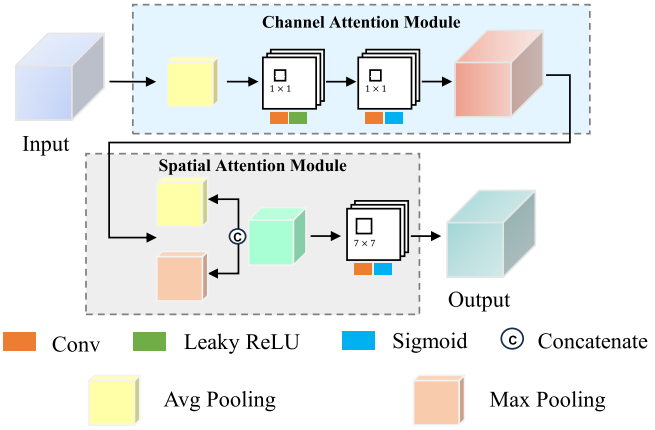


Fig. 2. Illustration of the CBAM structure.

feature responses by modeling interchannel dependencies through global pooling followed by two lightweight fully connected layers. The spatial attention submodule then emphasizes informative pixel regions by applying a convolutional operation over the aggregated channel descriptors.

This hierarchical attention design enables the network to selectively focus on the most discriminative spectral bands and spatial locations, thereby enhancing the accuracy and physical interpretability of the estimated abundances.

3) Abundance Generation: Finally, the attention-enhanced feature map $\boldsymbol{F}_{\text{cbam}}$ is passed through a fusion module $g_{\text{fusion}}(\cdot)$ to generate the initial abundance map. The module consists of two consecutive 1×1 convolutional layers that project the feature dimension from $2d$ to the number of endmembers R

$$\hat{\boldsymbol{A}}_{\text{init}} = g_{\text{fusion}}(\boldsymbol{F}_{\text{cbam}}) \tag{7}$$

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where $\hat{\boldsymbol{A}}_{\text{init}} \in \mathbb{R}^{R\times H\times W}$. To ensure that the estimated abundance maps satisfy the ANC and ASC, a softmax function is applied along the endmember dimension r at each spatial location (i, j)

$$\hat{\boldsymbol{A}}_{:,i,j} = \text{softmax}\left(\hat{\boldsymbol{A}}_{\text{init},:,i,j}\right) \tag{8}$$

which guarantees that for any pixel (i, j), the abundance vector $\boldsymbol{A}_{:,i,j} \in \mathbb{R}^{R\times 1}$ satisfies $\hat{A}_{r,i,j} \geq 0$ and $\sum_{r=1}^{R} \hat{A}_{r,i,j} = 1$.

Overall, the entire abundance estimation process can be concisely expressed as

$$\hat{\boldsymbol{A}} = \text{softmax}\left(g_{\text{fusion}}\left(\text{CBAM}\left(\text{Concat}\left[\, g_{\text{mlp}}(\boldsymbol{X}),\, g_{\text{cnn}}(\boldsymbol{X})\,\right]\right)\right)\right). \tag{9}$$

C. Endmember Estimation Module

The endmember estimation module aims to infer the spectral signatures of endmembers directly from the input hyperspectral image by first learning a global scene-level representation and then decoding the endmember matrix from it. It generates an endmember matrix $\hat{\boldsymbol{M}} = [\hat{\boldsymbol{m}}_1, \hat{\boldsymbol{m}}_2, \ldots, \hat{\boldsymbol{m}}_R] \in \mathbb{R}^{L\times R}$, where each column $\hat{\boldsymbol{m}}_r$ represents the spectral signature of the rth endmember. The process consists of two main stages: scene encoding and feature decoding.

1) Scene Encoding: The input hyperspectral image $\boldsymbol{X}$ is first processed by an encoder $g_{\text{enc}}(\cdot)$. The encoder consists of two 1×1 convolutional layers and projects the original L-dimensional spectral data into a d_e-dimensional latent feature space, producing a feature map $\boldsymbol{F}_{\text{enc}} \in \mathbb{R}^{d_e\times H\times W}$.

Global average pooling (GAP) is applied across the spatial dimensions to aggregate these features into a compact vector that captures the overall scene characteristics. The resulting $d_e \times 1 \times 1$ tensor is then flattened to produce the scene-level feature vector

$$z = \text{GAP}(\boldsymbol{F}_{\text{enc}}) \in \mathbb{R}^{d_e}. \tag{10}$$

2) Feature Decoding and Normalization: The global feature vector z is then fed into a decoder $g_{\text{dec}}(\cdot)$, which is implemented as a two-layer MLP. This decoder maps the compact feature vector back to the high-dimensional space of endmembers. Specifically, the MLP outputs a vector of size $L \times R$, which is then reshaped to form the initial endmember matrix. An l_2 normalization is subsequently applied along the spectral dimension to ensure that the estimated endmembers are represented based on their spectral shapes rather than magnitudes. The final endmember matrix $\hat{\boldsymbol{M}}$ is thus obtained as

$$\hat{\boldsymbol{M}} = \text{Normalize}\left(g_{\text{dec}}(z)\right). \tag{11}$$

Furthermore, to accelerate convergence and enhance physical interpretability, the bias term of the final layer in $g_{\text{dec}}(\cdot)$ is initialized using endmember signatures extracted by the classical VCA algorithm.

D. Spectral Reconstruction Module

The spectral reconstruction module aims to regenerate the hyperspectral image from the estimated endmember matrix $\hat{\boldsymbol{M}}$ and abundance maps $\hat{\boldsymbol{A}}$. Beyond the conventional LMM, this module is explicitly designed to capture complex and higher-order spectral interactions commonly observed in real-world scenes, thereby enhancing the fidelity of spectral reconstruction.

1) Linear Reconstruction Baseline: We first define a linear reconstruction baseline following the standard LMM formulation. The abundance tensor $\hat{\boldsymbol{A}}$ is reshaped into a matrix $\boldsymbol{A}' \in \mathbb{R}^{P\times N}$, and the reconstructed hyperspectral data are obtained as

$$\hat{\boldsymbol{X}}_{\text{lin}} = \hat{\boldsymbol{M}}\boldsymbol{A}' \in \mathbb{R}^{L\times N} \tag{12}$$

where $\hat{\boldsymbol{X}}_{\text{lin}}$ serves as the first-order approximation of the observed spectra.

2) Modeling High-Order Spectral Interactions: To overcome the limitations of the LMM, we recast nonlinear phenomena, such as multiple light scattering, as a structured progression of iterative refinements upon the linear reconstruction. Let $\boldsymbol{x}_j^{(i)} \in \mathbb{R}^L$ denote the spectral feature vector of the jth pixel at the ith interaction stage, with the initial state $\boldsymbol{x}_j^{(0)} = \hat{\boldsymbol{X}}_{\text{lin},:,j}$ representing the first-order linear response. The recursive generation of higher-order representations is formulated as

$$\boldsymbol{x}_j^{(i+1)} = \boldsymbol{x}_j^{(0)} \odot \left(\boldsymbol{W}^{(i)}\boldsymbol{x}_j^{(i)}\right) + \boldsymbol{x}_j^{(i)}, \quad j = 1, \ldots, N \tag{13}$$

where $\boldsymbol{W}^{(i)} \in \mathbb{R}^{L\times L}$ is a learnable weight matrix at the ith refinement level. The superscript i serves as a recursion index representing the order of nonlinear interactions. This process is applied up to a predefined maximum depth M, a hyperparameter that governs the highest order of nonlinear coupling. In the formulation, the term $(\boldsymbol{W}^{(i)}\boldsymbol{x}_j^{(i)})$ adaptively modulates interaction intensity, while the multiplicative term $\boldsymbol{x}_j^{(0)} \odot (, \ldots,)$ captures the band-wise coupling between the original spectrum and its evolving representation. The residual term $+\boldsymbol{x}_j^{(i)}$ ensures information preservation across iterations and stabilizes the recursive evolution.

The proposed formulation (13) is inspired by classical NLMMs while introducing a more expressive modeling mechanism. Traditional models, such as PPNM and MLM, rely on fixed physical interaction forms. Even when combined with neural networks, they typically estimate only a limited set of parameters within a rigid structure. In particular, MLM models nonlinear interactions through a recursive formulation in which all interaction orders are governed by a shared scalar parameter p_i. Different recursion stages, corresponding to higher-order interaction terms, are generated solely through the power expansion of this single parameter. As a result, all higher-order interactions remain coupled and cannot be independently modeled. Its extension, EMLM introduces band-wise variability by vectorizing this parameter, but still applies the same parameterization across recursion stages, lacking stage-wise flexibility. In contrast, the proposed formulation introduces a learnable matrix $\boldsymbol{W}^{(i)}$ at each recursion step, enabling stage-dependent and input-adaptive modulation of nonlinear interactions. This design decouples different interaction orders and significantly enhances the model's capacity to capture complex and higher-order mixing effects beyond predefined functional forms. Overall, by relaxing the rigid assumptions of predefined forms while preserving physical intuition, the proposed framework provides a compact and flexible approximation of complex multiscattering effects.

We further implement this recursive refinement using a stack of cross-network layers, denoted as $g_{\text{cross}}(\cdot)$, which naturally

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realizes the above formulation in an efficient and differentiable manner. The resulting nonlinear component is expressed as $\hat{\boldsymbol{X}}_{\text{nonlin}} = g_{\text{cross}}(\hat{\boldsymbol{X}}_{\text{lin}})$. Finally, the reconstructed hyperspectral image is obtained via a weighted fusion of the linear and nonlinear terms

$$\hat{\boldsymbol{X}} = \hat{\boldsymbol{X}}_{\text{lin}} + \alpha \hat{\boldsymbol{X}}_{\text{nonlin}} \tag{14}$$

where α is a learnable scalar that balances the linear reconstruction and the nonlinear interaction term. It is initialized with a small positive value to ensure that the reconstruction is initially dominated by the linear component, and is then optimized jointly with the network parameters through backpropagation, allowing the nonlinear contribution to be adaptively adjusted during training.

E. Training Strategy and Loss Function

To ensure stable convergence and improved unmixing performance, a stage-wise training strategy is adopted. Unlike conventional dual-stream or autoencoder-based methods that jointly optimize endmember extraction and abundance estimation, inaccurate endmember estimation in early training may introduce noisy gradients into the abundance branch, leading to error accumulation. To mitigate this issue, a progressive decoupled optimization scheme is employed. In the initial stage, the endmember estimation module is frozen, while the abundance estimation and reconstruction modules are updated, enabling the model to learn reliable abundance representations within a stable endmember space. In addition, VCA-extracted endmembers are incorporated as bias terms to provide a physically grounded initialization, which improves convergence stability. Subsequently, the model is fine-tuned in a joint training stage, where all modules are optimized end-to-end with stage-specific loss functions.

During the pretraining phase, the parameters of the endmember estimation module are frozen, while only the abundance estimation and spectral reconstruction modules are updated. The objective of this stage is to guide the network to learn the abundance distribution and perform nonlinear spectral reconstruction under a fixed endmember basis, thereby establishing a stable spectral–spatial representation. Over T iterations, the loss function for this phase is defined as

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{rec}} = \frac{1}{L \cdot N} \sum_{i=1}^{L} \sum_{j=1}^{N} \left(x_{i,j} - \hat{x}_{i,j}\right)^2 \tag{15}$$

where $x_{i,j}$ and $\hat{x}_{i,j}$ denote the spectral value at the ith band of the jth pixel in the original image $\boldsymbol{X}$ and its reconstruction $\hat{\boldsymbol{X}}$, respectively.

After the pretraining phase, the process proceeds to the second stage, referred to as the joint fine-tuning stage, which spans from iteration $T + 1$ until the end of training. In this stage, the parameters of the endmember estimation module are unfrozen, and the entire network is optimized in an end-to-end manner. The endmember representations $\hat{\boldsymbol{M}}$ are adaptively updated based on the reconstructed spectra, while the abundances $\hat{\boldsymbol{A}}$ and the reconstruction module are jointly adjusted. This coordinated optimization enables the network to achieve a globally optimal solution, ensuring that both endmembers and abundances converge to physically interpretable results.

To ensure the physical plausibility and accuracy of the unmixing results, several loss terms are introduced in this stage, including the reconstruction loss $\mathcal{L}_{\text{rec}}$, the minimum-volume constraint $\mathcal{L}_{\text{minvol}}$, the spectral angle distance (SAD) loss $\mathcal{L}_{\text{sad}}$, and the sparsity constraint $\mathcal{L}_{\text{sparse}}$. Their definitions are given below.

The reconstruction loss $\mathcal{L}_{\text{rec}}$, as in the first stage, is employed to preserve the fidelity of spectral reconstruction, and its formulation is given in (15).

The minimum-volume constraint $\mathcal{L}_{\text{minvol}}$ is derived from convex geometry theory, which states that all pixel spectra should lie within the simplex spanned by the estimated endmembers. To encourage purer endmember estimation, this constraint regularizes the estimated endmember matrix $\hat{\boldsymbol{M}} \in \mathbb{R}^{L \times R}$ such that the endmembers form a simplex with minimal volume. The formulation is given by

$$\mathcal{L}_{\text{minvol}} = \frac{1}{N \cdot R \cdot L} \sum_{i=1}^{R} \|\hat{\boldsymbol{m}}_i - \bar{\boldsymbol{m}}\|_F^2 \tag{16}$$

where $\hat{\boldsymbol{m}}_i$ denotes the ith estimated endmember, and $\bar{\boldsymbol{m}} = 1/R \sum_{i=1}^{R} \hat{\boldsymbol{m}}_i$ is their mean spectral vector.

To encourage the network to extract endmembers that are spectrally consistent with the initialized References, the SAD loss is employed. It measures the angular discrepancy between the estimated endmember matrix $\hat{\boldsymbol{M}}$ and the VCA-initialized matrix $\hat{\boldsymbol{M}}_{\text{vca}}$, ensuring spectral consistency between the two sets of endmembers. The loss $\mathcal{L}_{\text{sad}}$ is defined as the maximum SAD among all endmembers to maintain robustness under the worst case condition

$$\mathcal{L}_{\text{sad}} = \max_{r=1,\ldots,R} \left(\arccos \left(\frac{\hat{\boldsymbol{m}}_i^{\top} \hat{\boldsymbol{m}}_{\text{vca},i}}{\|\hat{\boldsymbol{m}}_i\|_2 \left\|\hat{\boldsymbol{m}}_{\text{vca},i}\right\|_2} \right) \right) \tag{17}$$

where $\hat{\boldsymbol{m}}_{\text{vca},i}$ denote the ith endmember vectors in $\hat{\boldsymbol{M}}_{\text{vca}}$.

In hyperspectral images, a pixel is typically composed of only a few constituent materials, which results in a sparse abundance matrix $\hat{\boldsymbol{A}} \in \mathbb{R}^{R \times N}$. To promote this sparsity, an $\ell_{1/2}$-norm regularization term is applied to promote sparsity in the abundance estimates. The sparsity loss is formulated as

$$\mathcal{L}_{\text{sparse}} = \frac{1}{R \cdot N} \sum_{i=1}^{R} \sum_{j=1}^{N} \sqrt{\left|\hat{a}_{i,j}\right| + \epsilon} \tag{18}$$

where $\hat{a}_{i,j}$ denotes the abundance of the ith endmember in the jth pixel, and ϵ is a constant introduced for numerical stability.

Finally, the overall loss function in the joint fine-tuning stage is formulated as a composite loss combining the aforementioned components

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{vol}} \mathcal{L}_{\text{minvol}} + \lambda_{\text{sad}} \mathcal{L}_{\text{sad}} + \lambda_{\text{sparse}} \mathcal{L}_{\text{sparse}} \tag{19}$$

where λ_{vol}, λ_{sad}, and λ_{sparse} are the hyperparameters controlling the contributions of the corresponding regularization terms. To provide a clear overview of the proposed unmixing process, the complete implementation procedure, including the two-stage training strategy and loss optimization, is summarized in Algorithm 1.

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Algorithm 1 Pseudo-Code of the Proposed Method

Input: hyperspectral image $\boldsymbol{X}$; hyperparameters $\lambda_{\text{vol}}, \lambda_{\text{sad}}, \lambda_{\text{sparse}}$; total epochs E; warm-up iterations T; nonlinear recursion number M.
Output: endmember estimation $\hat{\boldsymbol{M}}$, abundance maps $\hat{\boldsymbol{A}}$, reconstructed image $\hat{\boldsymbol{X}}$.

Extract initial endmembers using VCA to initialize the bias of the output layer in the endmember estimation branch;
Initialize the learnable scalar parameter α ;
Set $Epoch = 0$;
while $Epoch < E$ **do**
  $Epoch = Epoch + 1$;
  Estimate $\hat{\boldsymbol{A}}$ and $\hat{\boldsymbol{M}}$ using (9) and (11);
  Linear reconstruction is obtained using (12);
  Nonlinear refinement is iteratively performed for M steps using (13);
  $\hat{\boldsymbol{X}}$ is obtained by combining linear and nonlinear components using (14);
  if $Epoch \leq T$ **then**
    Compute loss $\mathcal{L}_{\text{stage1}}$ using (15);
  else
    Compute loss $\mathcal{L}_{\text{stage2}}$ using (19);
  end
  Update model parameters via backpropagation;
end

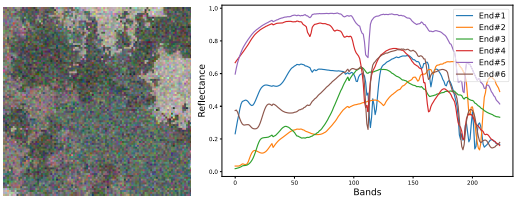


Fig. 3. RGB composite of the synthetic dataset and its corresponding reference endmembers.

IV. Experiments

The experimental section is organized as follows. First, eight state-of-the-art unmixing algorithms are introduced for comparison. Next, the four datasets used in our experiments are described in detail. The experimental parameters and evaluation metrics are then specified. Subsequently, we present the quantitative and visual results on one synthetic dataset and three real hyperspectral datasets. Finally, sensitivity analysis is conducted on the synthetic data, and ablation experiments are performed on the Jasper Ridge dataset to assess the contribution of each model component.

For comparison, eight representative unmixing methods were selected. Among them, the FCLSs method [10] and the multilinear mixing model (MLMp) [47] were included as traditional baselines. The remaining approaches are deep learning–based unmixing networks, including the cycle-consistency unmixing network (CyCU-Net) [24], the minimum simplex convolutional network (MiSiCNet) [48], the stationary wavelet convolutional network (SWCNet) [49], and the spatial–spectral adaptive fusion network (SSAFNet) [50]. In particular, SWCNet and SSAFNet represent recent

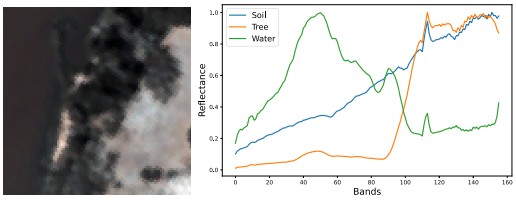


Fig. 4. RGB composite of the Samson dataset and its corresponding reference endmembers.

advances in hyperspectral unmixing. Furthermore, two nonlinear deep unmixing networks were considered, namely, the deep autoencoder-based augmented network (DAAN) [18] and MLM-1DAE [33]. For fairness, all methods requiring endmember initialization were initialized using the same VCA-derived endmembers.

A. Data Description

1) Synthetic Dataset: In constructing the synthetic dataset, six endmember spectra were randomly selected from the United States Geological Survey (USGS) digital spectral library, which contains 224 continuous spectral bands. Subsequently, an abundance matrix was generated using the Gaussian random field method provided by the hyperspectral image synthesis (HYDRA) toolbox. The abundance matrix has a dimension of $6 \times 10\,000$, corresponding to a spatial size of 100×100 pixels. Finally, the synthetic hyperspectral image was produced by mixing the endmembers and abundance matrix based on the MLM. An RGB representation of the synthetic hyperspectral image and the reference endmember maps are shown in Fig. 3.

2) Samson Dataset: In the experiments, a 95×95 pixel region from the Samson dataset was selected. Each pixel contains 156 spectral bands covering wavelengths from 401 to 889 nm. The scene mainly consists of three endmembers: soil, tree, and water. The RGB image of the Samson dataset and the corresponding reference endmember maps are shown in Fig. 4.

3) Jasper Ridge Dataset: The Jasper Ridge dataset is a hyperspectral image acquired in 1999 over the Jasper Ridge Biological Preserve by the Analytical Imaging and Geophysics (AIG) system. A subregion of 100×100 pixels was selected for the experiments. The image contains 224 spectral bands spanning wavelengths from 380 to 2500 nm. To mitigate the effects of water vapor absorption and atmospheric interference, several corrupted bands were removed, resulting in 198 valid bands. The dataset includes four main endmembers: road, soil, water, and tree. The RGB image of the Jasper Ridge dataset and the corresponding reference endmember maps are shown in Fig. 5.

4) Urban Dataset: The Urban dataset is a hyperspectral image acquired by the HYDICE sensor over an urban area in Copperas Cove, TX, USA. A 307×307 pixel region with a spatial resolution of 2×2 m^2 was selected for the experiments. After removing the corrupted bands affected by water vapor absorption and atmospheric interference, 162 valid spectral bands were retained. The scene mainly consists of four endmembers: asphalt, grass, tree, and roof. The RGB image of

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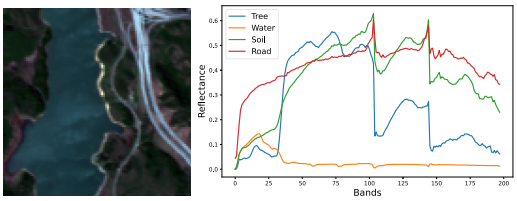


Fig. 5. RGB composite of the Jasper Ridge dataset and its corresponding reference endmembers.

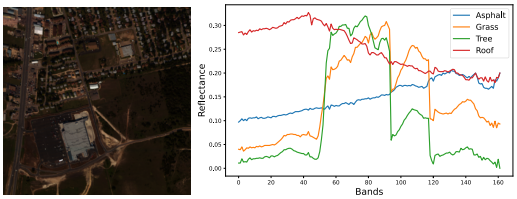


Fig. 6. RGB composite of the Urban dataset and its corresponding reference endmembers.

the Urban dataset and the corresponding reference endmember maps are shown in Fig. 6.

B. Experimental Settings

1) Evaluation Metrics: To quantitatively evaluate and compare the performance of the proposed method with other competing approaches, the root-mean-square error (RMSE) of the abundance estimation is adopted as the evaluation metric, which is defined as follows

$$\text{RMSE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{1}{R}\|\boldsymbol{a}_i - \hat{\boldsymbol{a}}_i\|^2} \tag{20}$$

where $\boldsymbol{a}_i$ and $\hat{\boldsymbol{a}}_i$ denote the reference and estimated abundance vectors of the ith pixel, respectively. In addition, the SAD is adopted as a quantitative metric for evaluating the accuracy of endmember estimation

$$\text{SAD} = \arccos\left(\frac{[\boldsymbol{m}_i, \hat{\boldsymbol{m}}_i]}{\|\boldsymbol{m}_i\| \cdot \|\hat{\boldsymbol{m}}_i\|}\right) \tag{21}$$

where $\boldsymbol{m}_i$ and $\hat{\boldsymbol{m}}_i$ denote the reference and estimated spectral signatures of the ith endmember, respectively.

2) Hyperparameter Settings: Based on extensive experiments, the hyperparameters for the four datasets were determined. The Adam optimizer was employed to update the network parameters. Specifically, the initial learning rate for the synthetic dataset was set to 0.001 with 650 training epochs, including 40 pretraining epochs. For the Samson and Jasper datasets, the learning rate was set to 0.002 with 280 training epochs, including 50 pretraining epochs, while for the Urban dataset, the learning rate was set to 0.002 with 150 training epochs, including 50 pretraining epochs. During training, the initial value of the parameter α in (14) was set to 0.05 for all datasets. For the regularization terms in (19), the volume coefficient λ_{vol} was set to 0.9, 0.5, 0.08, and 0.9 for the synthetic, Samson, Jasper, and Urban datasets, respectively. The spectral angle coefficient λ_{sad} was set to 0.005, 0.0002, 0.001, and 0.003, while the sparsity coefficient λ_{sparse} was set to 0.01, 0.001, 0.002, and 0.006 for the four datasets, respectively.

TABLE I
DETAILED CONFIGURATION OF THE MAJOR NETWORK MODULES

Module	Input	Output	Units / Channels
Spectral MLP	L	128	128
Spatial CNN	L	128	64
CBAM	256	256	-
Abundance Estimator	256	P	64
Endmember Estimator	L	$L \times P$	J
Spectral Reconstruction	L	L	Linear + Dropout

Moreover, the key input and output dimensions of major network modules are summarized in Table I. It is worth noting that the feature dimension J in the endmember estimation module differs slightly across datasets: $J = 64$ for the Jasper, Urban, and synthetic datasets, while $J = 128$ for the Samson dataset. In addition, the maximum recursion depth M is set to 4 in all experiments, which corresponds to the maximum value of the recursion index i in (13) and serves as a structural hyperparameter controlling the order of nonlinear interactions.

C. Experimental Results

1) Synthetic Dataset Dataset: The synthetic dataset was constructed following the method described in Section IV-A, where the MLM corresponds to (2). Table II presents the quantitative comparison results under a signal-to-noise ratio (SNR) of 10 dB, with the best results highlighted in bold. As observed, the proposed method achieves the lowest average SAD for endmembers, significantly outperforming other approaches in spectral estimation. Regarding the abundance estimation, although the proposed method yields the second-best RMSE, it maintains a highly competitive performance that is closely comparable to the top-performing baseline. Overall, the proposed framework demonstrates superior quantitative performance in the complex scenario. Furthermore, Figs. 7 and 8 illustrate the estimated abundance maps and endmember spectra obtained by the proposed method. The visual results clearly indicate that our method preserves spatial structures and restores spectral features more effectively than other competing algorithms, confirming its robustness and effectiveness in handling more challenging hyperspectral unmixing tasks.

2) Samson Dataset: On the Samson dataset, Fig. 9 presents the abundance maps obtained by different methods. As shown, the proposed method produces abundance distributions that are most consistent with the reference maps. The endmember visualization in Fig. 10 further demonstrates the effectiveness of the proposed method in accurately estimating spectral signatures. The quantitative results are summarized in Table III. The proposed method achieves the best performance in terms of abundance RMSE, while ranking second in endmember SAD, slightly behind SWCNet. A detailed analysis of the individual endmembers shows that our approach achieves the lowest SAD for tree and soil, but performs slightly worse for water. This may be attributed to the low reflectance and high spectral variability of water pixels [51], which make their spectral signatures more difficult to model accurately. This phenomenon is commonly observed in hyperspectral unmixing studies, as water spectra are known to be sensitive to illumination and noise.

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TABLE II

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE SYNTHETIC DATASET, SHOWING SAD FOR ENDMEMBER ESTIMATION AND RMSE FOR ABUNDANCE ESTIMATION. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Methods		FCLSs	MLMp	CyCU-Net	DAAN	MiSiCNet	SWCNet	SSAFNet	MLM-1DAE	Proposed
SAD	End#1	0.0874	0.2903	0.4943	0.0890	0.0810	0.0820	0.0656	0.1965	**0.0632**
	End#2	0.0950	0.3876	0.1134	0.0917	0.1370	**0.0704**	0.1062	0.2387	0.0780
	End#3	0.0915	0.3995	0.2994	0.0967	0.0729	0.0718	0.0728	0.1591	**0.0617**
	End#4	0.0896	0.3187	0.0608	0.0875	0.0602	0.0595	**0.0431**	0.1716	0.0480
	End#5	0.0618	0.1777	0.1683	0.0619	0.0955	0.0628	0.0816	**0.0520**	0.0693
	End#6	0.1164	0.3610	**0.0521**	0.1194	0.0674	0.0590	0.0558	0.2748	0.0714
Mean SAD		0.0903	0.3225	0.1980	0.0910	0.0857	0.0676	0.0708	0.1821	**0.0653**
RMSE		0.0899	0.0910	0.1420	0.1098	**0.0664**	0.0714	0.0720	0.1440	0.0689

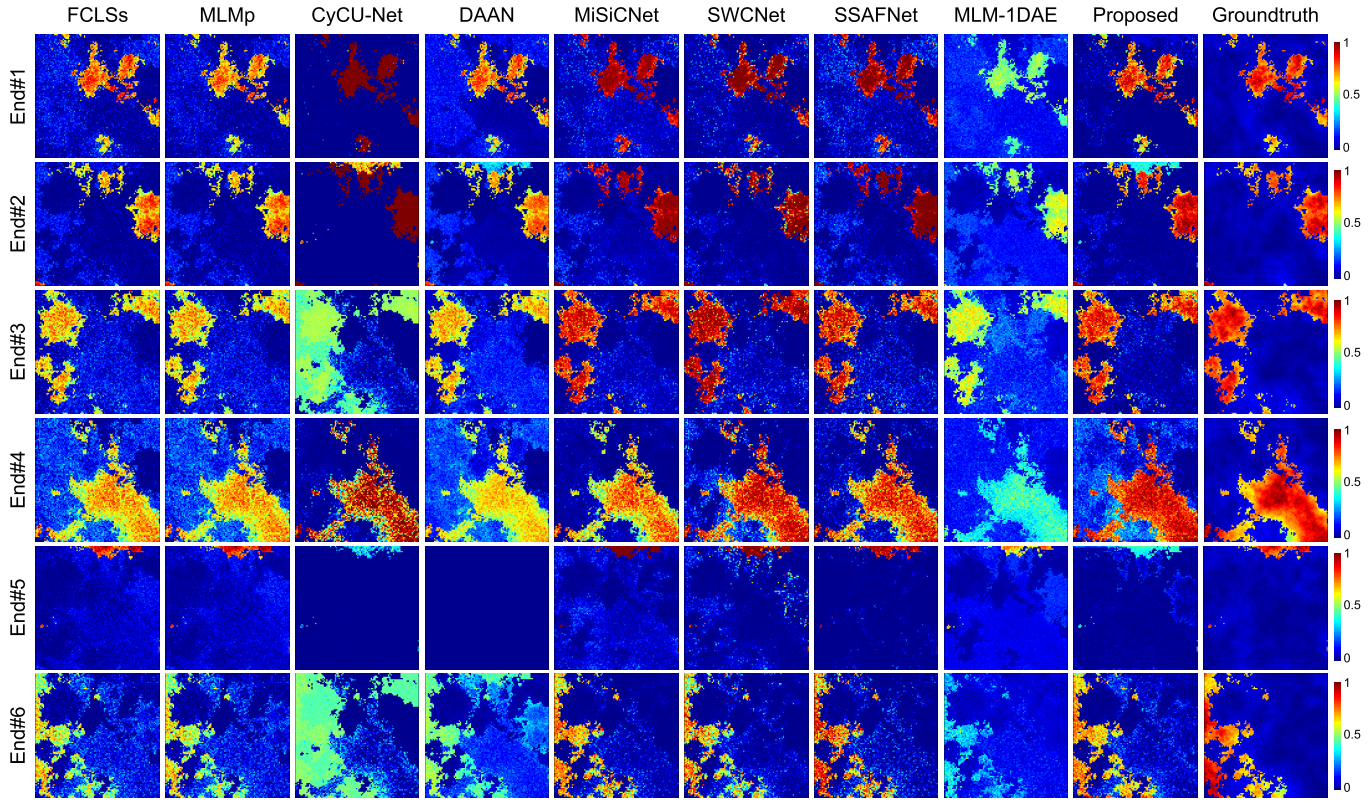


Fig. 7. Abundance estimation maps of different methods on the Synthetic dataset. The methods are arranged from left to right: FCLSs, MLMp, CyCU-Net, DAAN, MiSiCNet, SWCNet, SSAFNet, MLM-1DAE, the proposed method, and the ground truth.

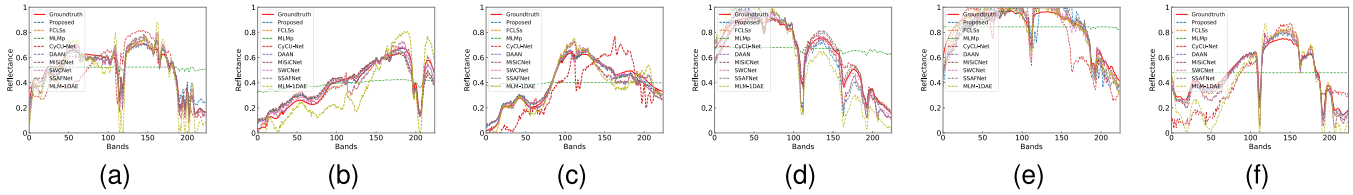


Fig. 8. Endmember estimation results of different methods on the synthetic dataset. (a) End#1. (b) End#2. (c) End#3. (d) End#4. (e) End#5. (f) End#6.

3) Jasper Ridge Dataset: As illustrated in Figs. 11 and 12, the proposed method produces abundance and endmember estimates of high visual quality on the Jasper dataset. The abundance maps show that our method is capable of recovering finer spatial details. In comparison, several competing approaches exhibit less satisfactory separation of the road region, while more recent methods such as SWC-Net and SSAF-Net achieve improved performance but still miss certain structural details. As highlighted by the red boxes in Fig. 13, the proposed method yields abundance maps that more closely match the reference, particularly in challenging road areas. Regarding endmember estimation, the proposed method yields results that are highly consistent with the reference signatures. As summarized in Table IV, our method attains the best overall performance in both SAD and RMSE. Although its average endmember SAD is close to that of SSAF-Net, the proposed approach achieves slightly higher accuracy, particularly for the water endmember. This improvement is attributed to the

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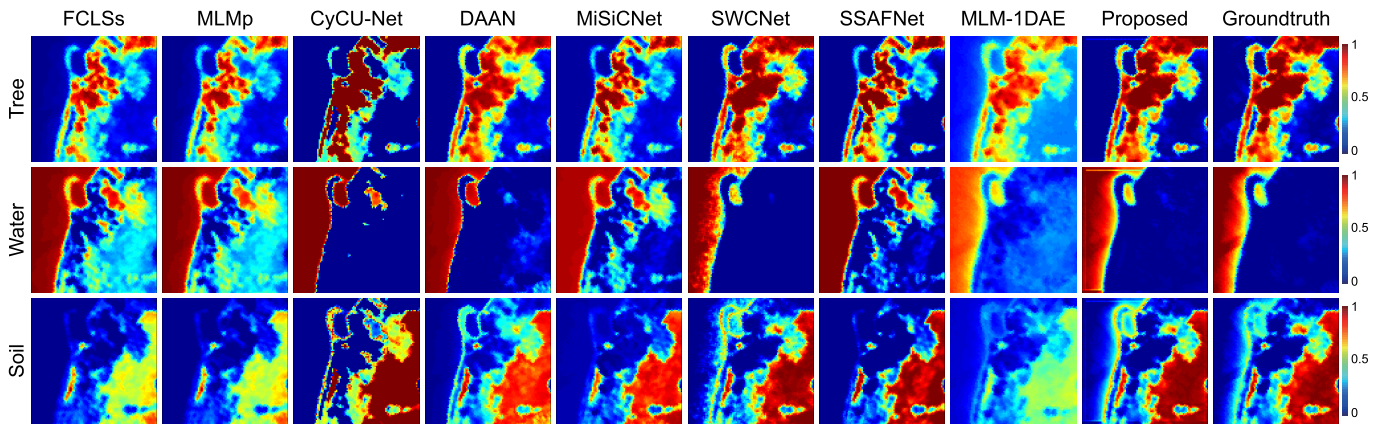


Fig. 9. Abundance estimation maps of different methods on the Samson dataset. (Top to bottom) Tree, water, and soil. The methods are arranged from left to right: FCLSs, MLMp, CyCU-Net, DAAN, MiSiCNet, SWCNet, SSAFNet, MLM-1DAE, the proposed method, and the ground truth.

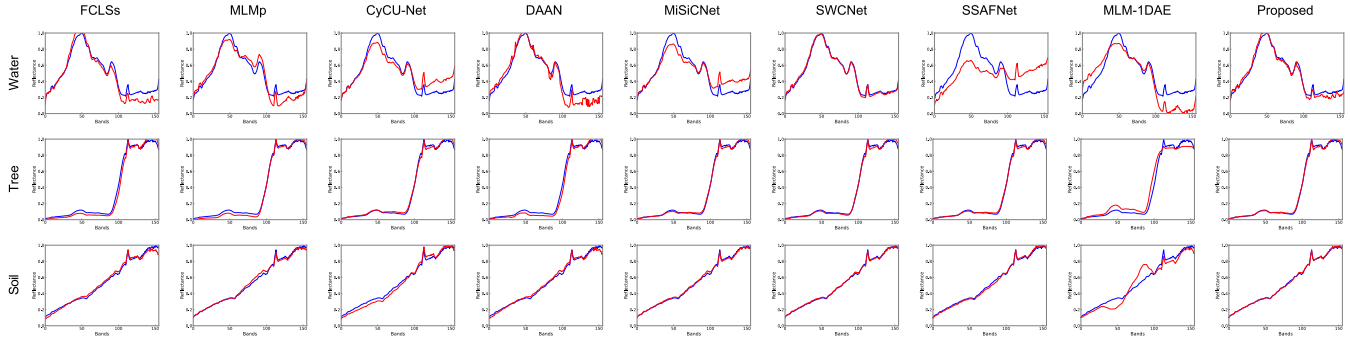


Fig. 10. Endmember estimation results of different methods on the Samson dataset. Blue lines indicate the ground reference, while red lines indicate the estimated endmembers. (Top to bottom) Water, tree, and soil. The methods are arranged from left to right: FCLSs, MLMp, CyCU-Net, DAAN, MiSiCNet, SWCNet, SSAFNet, MLM-1DAE, the proposed method, and the ground truth.

TABLE III

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE SAMSON DATASET, SHOWING SAD FOR ENDMEMBER ESTIMATION AND RMSE FOR ABUNDANCE ESTIMATION. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Methods		FCLSs	MLMp	CyCU-Net	DAAN	MiSiCNet	SWCNet	SSAFNet	MLM-1DAE	Proposed
SAD	Soil	0.0404	0.0315	0.0515	0.0362	0.0200	0.0130	0.0242	0.1201	**0.0110**
	Tree	0.0713	0.0453	0.0350	0.0698	0.0285	0.0309	0.0277	0.1093	**0.0271**
	Water	0.1304	0.1208	0.1661	0.1465	0.1710	**0.0269**	0.4032	0.2664	0.0694
Mean SAD		0.0807	0.0659	0.0842	0.0842	0.0732	**0.0236**	0.1517	0.1653	0.0358
RMSE		0.2289	0.2329	0.1256	0.0776	0.1618	0.0310	0.1465	0.1759	**0.0283**

SAD regularization term in the endmember estimation loss, which constrains the maximum deviation between estimated and reference endmembers. Such a mechanism promotes more balanced optimization across all endmembers, thereby enhancing the stability and overall fidelity of the estimated spectral signatures.

4) Urban Dataset: To further evaluate the performance of the proposed method in complex real-world environments, experiments were also conducted on the Urban dataset. The experimental results are presented in Table V. For abundance estimation, the proposed method achieves the lowest RMSE among all competing methods, demonstrating its superior capability in recovering abundance distributions. This advantage is also evident in the abundance maps shown in Fig. 14, where the estimated abundances exhibit high consistency with the reference maps. For endmember estimation, although the proposed method does not outperform SWCNet in terms of overall SAD, it remains highly competitive with the other methods. The relatively larger error mainly stems from the roof endmember, while the estimated spectra for the remaining endmembers show good agreement with the reference spectra shown in Fig. 15.

D. Discussion

1) Robustness Analysis: To evaluate the robustness of the proposed method against noise, experiments are conducted under different SNRs, and the corresponding abundance estimation and endmember estimation results are illustrated in Fig. 16. The left plot shows the RMSE of abundance estimation, while the right plot presents the SAD of endmember estimation. As shown in the figure, the proposed method maintains relatively stable abundance estimation performance across different noise levels, with only minor variations in RMSE as the noise intensity changes. Meanwhile, the endmember estimation performance shows a clear improvement as the noise level decreases, with the SAD gradually decreasing

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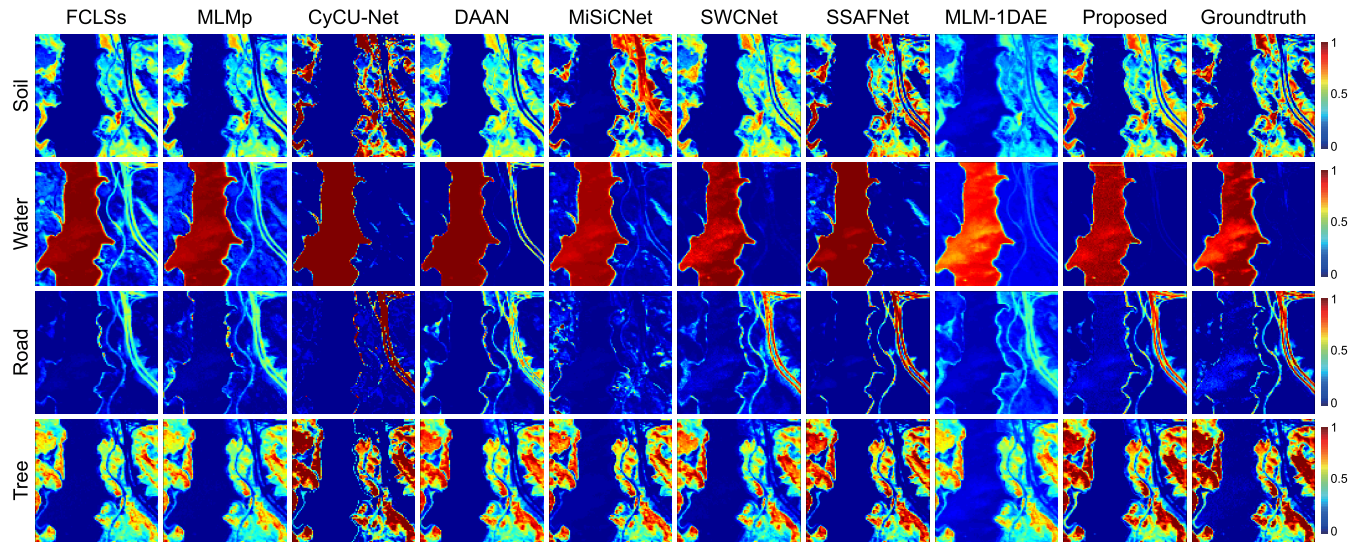


Fig. 11. Abundance estimation maps of different methods on the Jasper dataset. (Top to bottom) Soil, water, road, and tree. The methods are arranged from left to right: FCLSs, MLMp, CyCU-Net, DAAN, MiSiCNet, SWCNet, SSAFNet, MLM-1DAE, the proposed method, and the ground truth.

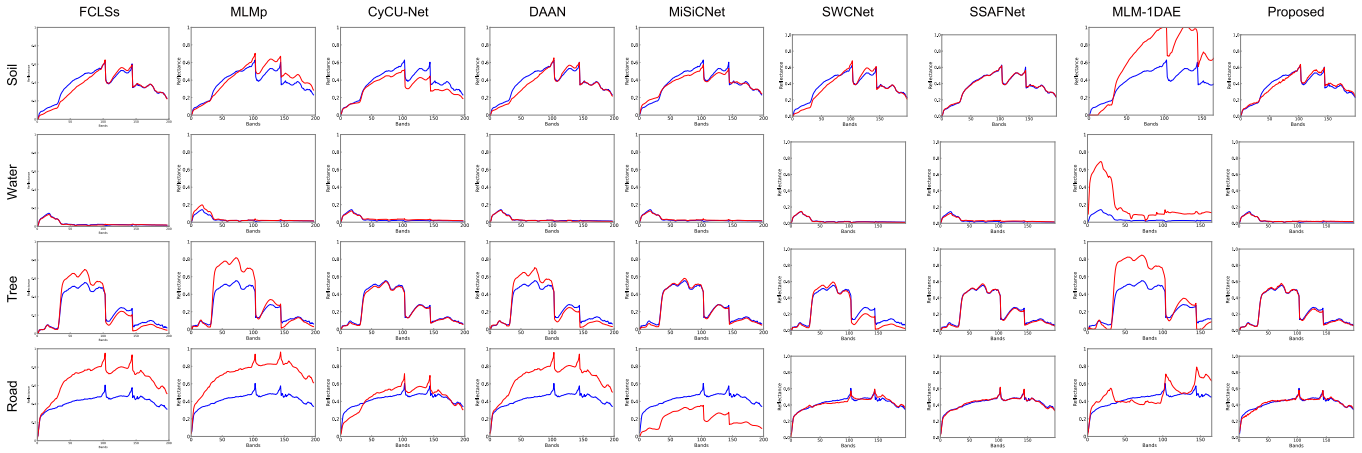


Fig. 12. Endmember estimation results of different methods on the Jasper dataset. Blue lines indicate the ground reference, while red lines indicate the estimated endmembers. (Top to bottom) Soil, water, tree, and road. The methods are arranged from left to right: FCLSs, MLMp, CyCU-Net, DAAN, MiSiCNet, SWCNet, SSAFNet, MLM-1DAE, the proposed method, and the ground truth.

TABLE IV

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE JASPER DATASET, SHOWING SAD FOR ENDMEMBER ESTIMATION AND RMSE FOR ABUNDANCE ESTIMATION. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Methods		FCLSs	MLMp	CyCU-Net	DAAN	MiSiCNet	SWCNet	SSAFNet	MLM-1DAE	Proposed
SAD	Tree	0.1481	0.1265	0.0258	0.1475	0.0495	0.1574	**0.0375**	0.1672	0.0406
	Water	0.0893	0.1032	0.2221	0.0964	0.1439	**0.0768**	0.2290	0.1356	0.1390
	Soil	0.1166	0.1253	**0.0641**	0.1224	0.0696	0.1117	0.0195	0.1270	0.0874
	Road	0.0901	0.0569	0.1295	0.0866	0.3200	0.0520	**0.0231**	0.2104	0.0410
Mean SAD		0.1110	0.1030	0.1104	0.1132	0.1458	0.0995	0.0773	0.1600	**0.0770**
RMSE		0.1241	0.1169	0.0787	0.0945	0.1155	0.0788	0.0584	0.1383	**0.0382**

from low to high SNRs. Compared with the competing methods, the proposed method maintains competitive performance across the tested noise levels, particularly in endmember estimation.

2) Sensitivity Analysis: To evaluate the robustness of the proposed model under the 6-endmember scenario, a sensitivity analysis was conducted on the synthetic dataset. A grid-search strategy was used to perturb the learning rate and the weighting coefficients in (19). The results are shown in Fig. 17, where the left and right vertical axes represent the abundance RMSE and the endmember SAD, respectively. As shown in Fig. 17, the model exhibits different sensitivities to various hyperparameters. In particular, the method is robust to the volume regularization term λ_{vol}, with RMSE varying

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TABLE V

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE URBAN DATASET, SHOWING SAD FOR ENDMEMBER ESTIMATION AND RMSE FOR ABUNDANCE ESTIMATION. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Methods		FCLSs	MLMp	CyCU-Net	MiSiCNet	SWCNet	SSAFNet	MLM-1DAE	Proposed
SAD	Asphalt	0.2095	0.0890	0.0980	0.2802	0.1063	0.1359	0.2367	**0.0727**
	Grass	0.4133	0.8826	0.2886	1.3180	0.0781	0.1901	0.4593	**0.0698**
	Tree	0.2894	0.1293	0.2371	0.1231	**0.0416**	0.0909	0.1005	0.1954
	Roof	0.8228	0.5713	0.1495	**0.1088**	0.1723	0.6474	0.2473	0.4726
Mean SAD		0.4338	0.4180	0.1933	0.4575	**0.0996**	0.2661	0.2610	0.2026
RMSE		0.1682	0.2749	0.2853	0.2703	0.1297	0.2693	0.1969	**0.0827**

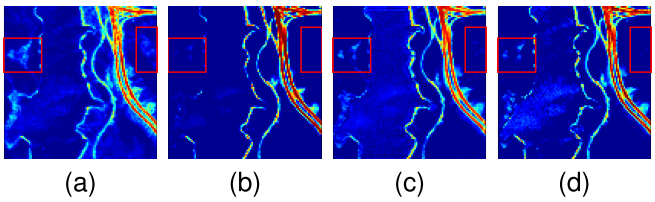


Fig. 13. Abundance maps of selected methods on the road region, with red boxes highlighting challenging areas. (a) SWCNet, (b) SSAFNet, (c) proposed method, and (d) ground truth.

TABLE VI

ABLATION STUDY OF THE PROPOSED MODEL ON THE JASPER DATASET

Description	RMSE	SAD
Nonlinear interaction module removed	0.0480	0.0826
Recursive nonlinear interaction replaced by 2-layer MLP	0.0661	0.1125
Spectral branch only (without spatial modeling)	0.0543	0.0811
Spatial branch only (without spectral modeling)	0.1143	0.1073
CBAM-based spectral–spatial fusion removed	0.1739	0.2546
VCA-based endmember initialization removed	0.1209	0.3007
Pretraining stage removed	0.0572	0.0866
Minimum-volume regularization removed	0.0814	0.1073
Endmember SAD regularization removed	0.0671	0.1740
Abundance sparsity regularization removed	0.0858	0.1284
Full proposed model (baseline)	**0.0382**	**0.0770**

slightly between 0.0689 and 0.0726 and SAD remaining below 0.0729. In contrast, the sparsity regularization λ_{sparse} has a more significant impact, where deviations from the optimal value lead to noticeable degradation in both RMSE and SAD. Moreover, the optimization trends of abundance and endmembers are not always consistent. For example, as the learning rate or λ_{vol} increases, the RMSE may increase while the SAD decreases. This indicates a tradeoff between abundance accuracy and spectral consistency. In this work, priority is given to optimizing abundance estimation performance. Overall, the proposed method demonstrates stable performance across a wide range of hyperparameter settings, with optimal results achieved by balancing the regularization terms.

3) Ablation: To evaluate the effectiveness of the proposed model across its different modules and regularization terms, a systematic ablation study was conducted on the Jasper dataset. The experiments cover the abundance estimation module, the endmember estimation module, the nonlinear reconstruction module, as well as the regularization terms in (19). All results are compared with the baseline performance of the full model, as summarized in Table VI.

For the nonlinear reconstruction module, removing the nonlinear component leads to an increase in both RMSE and SAD, indicating that nonlinear modeling contributes positively to unmixing accuracy. Replacing the cross-network with a standard two-layer MLP further reduces performance, demonstrating that the designed cross-network more effectively captures the complex nonlinear relationships inherent in mixed pixels.

Within the abundance estimation module, ablating either the spectral or spatial branch results in decreased estimation accuracy. Notably, retaining only the spatial branch leads to more pronounced degradation, suggesting that spectral information plays a dominant role in abundance estimation for this dataset, while the combination of spectral and spatial features remains crucial. Furthermore, removing the CBAM module causes a substantial drop in performance, confirming the importance of adaptive feature selection in suppressing redundant features and enhancing relevant ones.

In the endmember estimation module, removing the VCA-based initialization leads to increases in both SAD and abundance RMSE, underscoring the importance of proper initialization in stabilizing endmember estimation and promoting convergence. Removing the 50-epoch pretraining stage also results in performance degradation, although its effect is relatively less pronounced.

Regarding the loss function regularization, removing the minimum-volume constraint, the endmember SAD term, or the abundance sparsity term each leads to a reduction in unmixing accuracy. Among these, excluding the endmember SAD term has the most pronounced effect on endmember estimation, demonstrating its critical role in improving spectral consistency.

Furthermore, to provide additional insight into the learning behavior of the proposed nonlinear reconstruction module, the evolution of the learnable parameter α in (14) was analyzed on the Samson dataset, as shown in Fig. 18. Starting from the initial value of 0.05, α gradually decreases throughout the training process and finally converges to approximately 0.02. A slight fluctuation can be observed around the phase boundary between the pretraining and joint-training stages, after which α continues to decrease and gradually stabilizes. This evolution illustrates the adaptive adjustment of the nonlinear reconstruction component during the optimization process.

Overall, the ablation results confirm that nonlinear modeling, the spectral–spatial dual-branch structure, the CBAM module, VCA-based initialization, and the multiple regularization terms all contribute to the model's performance. The components act complementarily, and the full model achieves optimal unmixing results only when all modules and regularization terms are simultaneously employed.

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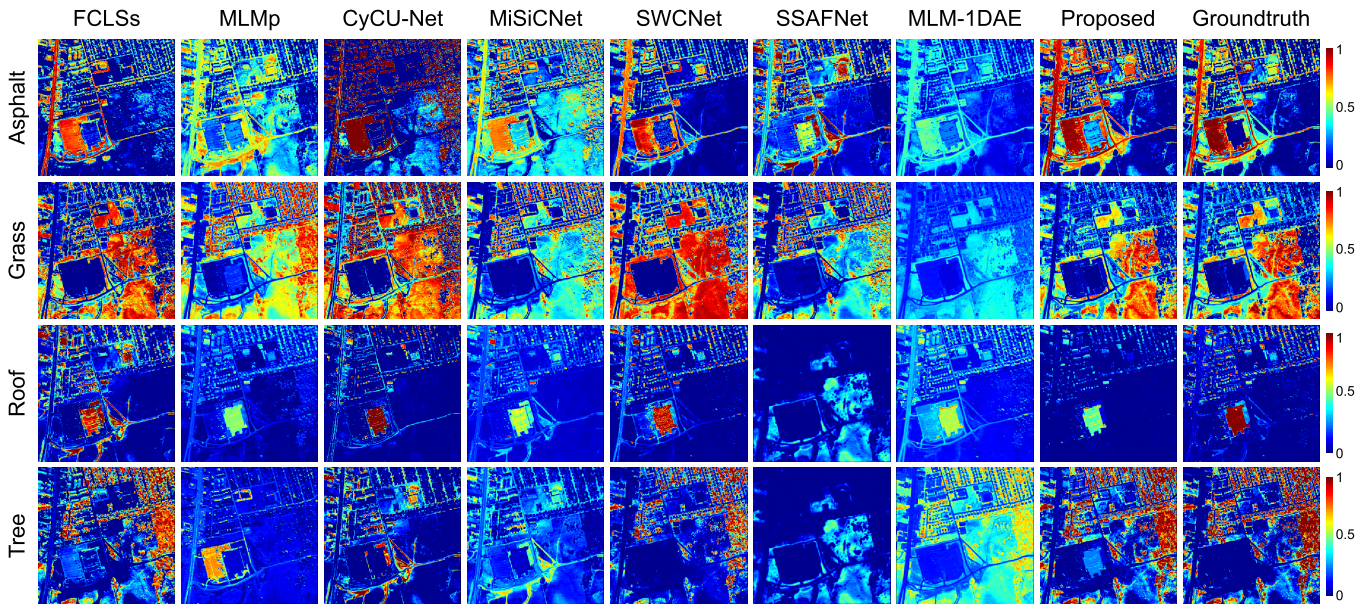


Fig. 14. Abundance estimation maps of different methods on the urban dataset. The methods are arranged from left to right: FCLSs, MLMp, CyCU-Net, MiSiCNet, SWCNet, SSAFNet, MLM-1DAE, the proposed method, and the ground truth.

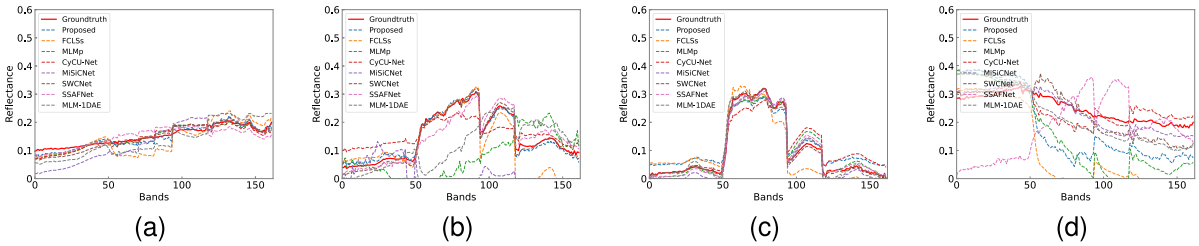


Fig. 15. Endmember estimation results of different methods on the urban dataset. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof.

TABLE VII

COMPUTATIONAL TIME IN SECONDS (S) FOR DIFFERENT METHODS ON THE SYNTHETIC, SAMSON, AND JASPER DATASETS

Methods	FCLSs	MLMp	CyCU-Net	DAAN	MiSiCNet	SWCNet	SSAFNet	MLM-1DAE	Proposed
Synthetic	6.25	42.16	10.51	1072.97	632.18	76.86	1356.51	614.17	323.91
Samson	1.73	11.25	13.36	323.95	550.53	10.01	35.55	227.05	104.32
Jasper	2.26	6.22	4.65	382.70	522.29	10.13	71.16	147.06	163.20
Urban	62.79	73.82	288.27		835.75	1928.74	3825.15	2129.27	1043.83

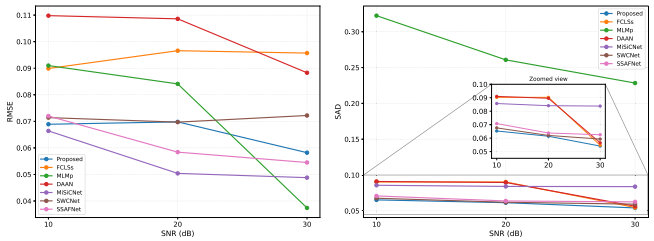


Fig. 16. Abundance and endmember estimation performance under different noise levels.

E. Computational Cost

To ensure a fair comparison, all experiments were conducted on the same hardware platform equipped with an Intel Core i5-8265U CPU (1.6 GHz) and 8-GB RAM. The proposed model was implemented using Python 3.11.5 and PyTorch 2.1.2. Table VII reports the runtime of all competing methods across the four datasets. As expected, deep learning-based approaches generally require more computation time than traditional machine learning methods due to their model complexity. Nevertheless, the proposed method achieves competitive efficiency within the deep learning category, exhibiting shorter runtime than most learning-based baselines while maintaining superior unmixing performance. This demonstrates that the proposed framework offers a favorable balance between computational cost and accuracy.

The computational complexity of the proposed method is primarily determined by three modules: abundance estimation, endmember estimation, and reconstruction. Following the notations in Section III, the abundance estimation module incurs a complexity of $\mathcal{O}(NLd)$, while the endmember estimation entails $\mathcal{O}(LR + NL)$. For the reconstruction module, the linear mixing between $\hat{\boldsymbol{M}}$ and $\hat{\boldsymbol{A}}$ contributes $\mathcal{O}(NLR)$, whereas the M-step recursive nonlinear refinement in (13) results in $\mathcal{O}(MNL^2)$. Consequently, the overall computational complexity is $\mathcal{O}(NLR + MNL^2 + NLd)$. Notably, the nonlinear

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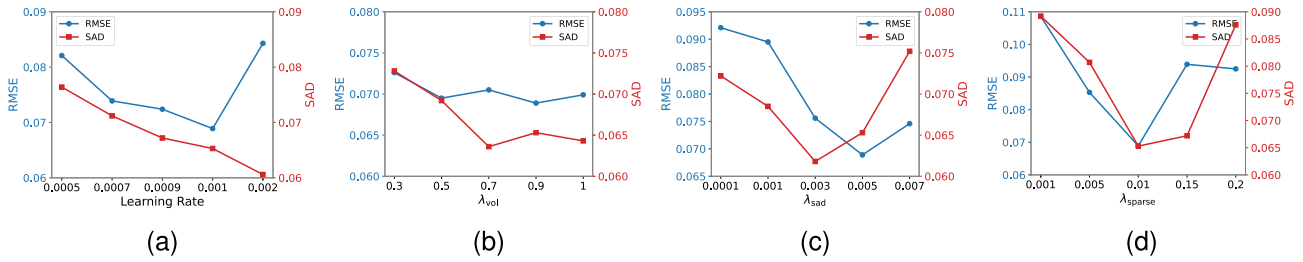


Fig. 17. RMSE as functions of the parameters learning rate, on the synthetic dataset. (a) Learning rate, (b) λ_{vol}, (c) λ_{sad}, and (d) λ_{sparse}.

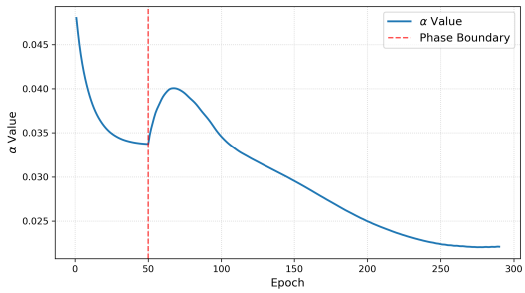


Fig. 18. Evolution of the learnable parameter α during training on the Samson dataset.

term $\mathcal{O}(MNL^2)$ dominates the total cost due to its iterative refinement process.

V. Conclusion

This work introduces a decoupled endmember–abundance framework for hyperspectral unmixing, incorporating targeted designs in feature extraction, endmember modeling, and nonlinear reconstruction. For abundance estimation, the proposed model fully exploits both spectral and spatial information through a cascaded attention mechanism. Specifically, spectral attention first identifies the most discriminative channel-wise feature combinations, while spatial attention subsequently locates the most informative regions. This design yields a refined and information-dense feature representation, which is then processed by convolutional layers to produce accurate abundance estimates. For endmember estimation, the model applies a 1×1 convolution for dimensionality reduction, followed by pooling to capture global contextual cues, and a fully connected network to estimate the endmember signatures. To enhance stability and interpretability, VCA-derived endmembers are incorporated as bias initialization in the final layer, providing a more favorable starting point for optimization. In the reconstruction stage, we propose a novel high-order scattering formulation inspired by physical interaction mechanisms, whose parameters are learned in a data-driven manner. In this formulation, raw features are progressively combined with interaction-enriched composite features via multiplicative operations. This mechanism systematically builds higher-order nonlinear relationships and is implemented using the structure of cross networks. The final reconstructed spectrum is obtained by aggregating both linear and nonlinear components, yielding a more physically faithful representation of hyperspectral mixing processes. Experimental results demonstrate that the proposed method consistently outperforms representative state-of-the-art approaches in terms of endmember accuracy, abundance estimation, and overall reconstruction error, thereby validating the effectiveness of each module as well as the superiority of the overall framework.

Despite its effectiveness, the proposed method still has some limitations. In particular, its performance depends on the quality of the initial endmember estimation, and the current recursive structure does not explicitly account for spectral variability or adaptive depth. To address these issues, future research will focus on two directions. First, we aim to incorporate endmember bundles to better model spectral variability caused by illumination and topography. Second, an adaptive stopping mechanism will be explored to dynamically adjust the recursion depth based on pixel complexity, thereby further improving the balance between efficiency and accuracy.

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